

ROYAL FITNESS CLUB

Nutrition & Healthy Eating Certificate

MODULE 5

Metabolism & Energy Balance

Duration	45 minutes
Course Code	CS_B2
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: What Is Metabolism?

Metabolism — The Sum of All Chemical Reactions

Metabolism refers to the entirety of chemical reactions that occur within living cells to maintain life. For body composition and fitness purposes, we focus on energy metabolism — the processes by which nutrients are converted to ATP and heat.

Metabolic Process	Category	Function	Hormonal Control
Glycolysis	Catabolism	Glucose → Glycogen (storage)	Glucagon (promotes), Insulin (promotes via glucose uptake)
Beta-oxidation	Catabolism	Fatty acids → Acetyl-CoA	Glucagon stimulate; insulin inhibits
Amino acid catabolism	Catabolism	Amino acids → glucose (gluconeogenesis)	Glucagon promotes; insulin inhibits
Glycogen synthesis	Anabolism	Glucose → glycogen (storage)	Insulin promotes
Triglyceride synthesis	Anabolism	Fatty acids + glycerol → stored fat	Insulin promotes; caloric surplus drives
Protein synthesis	Anabolism	Amino acids → muscle protein	Insulin + mTORC1 + resistance training

BMR — What Determines Your Metabolic Rate

Basal Metabolic Rate (BMR) is determined by multiple factors, some modifiable and some not:

Factor	Influence on BMR	Modifiable?
Lean muscle mass	Each kg muscle burns ~13 kcal/day at rest	Yes (resistance training increases muscle → increases BMR)
Body size (height, weight)	Larger bodies have higher absolute BMR	Partially
Age	BMR decreases ~1–2% per decade	Partially (muscle preservation reduces decline)
Sex	Males average 5–10% higher BMR (more muscle, less fat)	No (biological)
Thyroid function	Thyroid hormones (T3, T4) regulate metabolic rate	Yes (with medical treatment if impaired)
Genetics	Individual variation in mitochondrial efficiency	No
Temperature adaptation	Living in cold climate increases BMR slightly	No (environment-dependent)

The most effective way to increase BMR: Build lean muscle mass through resistance training. A 5kg increase in lean mass increases resting metabolic rate by approximately 65 kcal/day — not massive in isolation, but compounding over years plus NEAT and exercise activity creates substantial impact.

SECTION 2: Metabolic Adaptation — Why Diets Fail

Metabolic Adaptation: The Diet Plateau Explained

Metabolic adaptation (sometimes called "adaptive thermogenesis") is the body's multi-layered defence against weight loss. When you reduce calories, the body responds with mechanisms to conserve energy and defend body weight — a survival system evolved over millions of years.

Adaptation	Mechanism	Magnitude	Timeframe
Loss of RMR and thyroid hormone reduction (T4→T3 conversion ↓)		100 to 300 kcal/day	2–4 weeks
Unconscious reduction in fidgeting, posture activity, spontaneous activity		100 to 400 kcal/day	1–2 weeks
TEF reduction	Less food → less energy to digest food	50 to –100 kcal/day	Immediate
Hunger (ghrelin ↑); leptin (satiety ↓)	Significantly increases hunger		2–4 weeks
Exercise efficiency	Distance run at lower calorie cost (mass ↓)	50 to 150 kcal/day	Gradual

Total metabolic adaptation: After significant weight loss (10%+ body weight), metabolic adaptation can reduce TDEE by 300–500 kcal/day below what equations would predict for the new body weight. This is why "The Biggest Loser" contestants regain weight — their metabolism is chronically suppressed.

Strategies to Minimise Metabolic Adaptation

- Moderate deficit only: -500 kcal maximum; never go below 1200 kcal (F) or 1500 kcal (M)
- High protein: preserves lean mass, preventing RMR loss from muscle catabolism
- Resistance training: maintains or increases lean mass during deficit
- Diet breaks: 1–2 weeks at maintenance calories every 8–12 weeks of dieting (restores leptin, reduces ghrelin)
- Refeed days: 1–2 days per week at maintenance (smaller version of diet break)
- High NEAT: deliberately maintain non-exercise movement (steps target: 8000–10,000/day)

SECTION 3: Insulin — The Master Partitioning Hormone

Insulin: Far More Than "Fat Storage"

Insulin is a peptide hormone secreted by pancreatic beta cells in response to elevated blood glucose. It is often demonised in popular culture as "the fat storage hormone," but this is a profound oversimplification.

What insulin actually does:

- Drives glucose into muscle cells (via GLUT4 translocation) — muscle glycogen synthesis

- Drives glucose into adipose tissue — fat storage when in surplus
- Drives amino acids into muscle cells — potent MPS activator (synergistic with leucine)
- Inhibits gluconeogenesis (liver stops making glucose from amino acids)
- Inhibits lipolysis (fat breakdown) in adipose tissue
- Activates glycogen synthase — glycogen storage in liver and muscle
- Promotes protein synthesis via PI3K/Akt pathway

The truth about insulin and fat: Insulin promotes fat storage only in the context of caloric surplus. In a caloric deficit, fat is mobilised even with normal insulin levels because fat oxidation is driven by energy demand, not insulin alone. You can eat high-carb and lose fat in a deficit; you can eat low-carb and gain fat in a surplus. Energy balance remains primary.

Insulin Resistance: Prevention and Reversal

Mechanism	Cause	Prevention/Treatment
GLUT4 downregulation	Chronic hyperinsulinaemia; exercise (activates GLUT4 independently of insulin)	Weight loss
Ectopic fat deposition	Caloric surplus; especially fructose-driven hepatic lipogenesis	Reduced refined sugar; omega-3
Chronic inflammation	Adipose tissue macrophage infiltration; elevated TNF- α	Mediterranean Diet; weight loss; omega-3; exercise
Sleep deprivation	Cortisol increase $\rightarrow$ insulin receptor resistance	8+ hours sleep; sleep hygiene protocol
Physical inactivity	Reduced muscle glucose uptake	150+ min/week moderate exercise; reduce sitting

Module 5 (CS_B2) Key Takeaways

- Metabolism is the totality of chemical reactions in cells; energy metabolism determines body composition
- BMR is primarily determined by lean muscle mass — resistance training increases metabolic rate
- Metabolic adaptation reduces TDEE by 300–500 kcal/day after significant weight loss; diet breaks help restore it
- Insulin is anabolic and fat-storing — but fat is gained only in a caloric surplus, not from carbs per se
- NEAT is the most variable and underrated component of TDEE; 8000–10,000 steps/day burns 200–400 kcal extra
- Insulin resistance is driven by excess adipose tissue, inflammation, and inactivity — all modifiable

1. What is adaptive thermogenesis and how much can it reduce expected TDEE after weight loss?

2. Explain the role of leptin and ghrelin in the metabolic adaptation response to dieting.
3. Why is insulin not simply "the fat storage hormone" and how does exercise modify its effects?
4. What is a "diet break" and mechanistically why does it help overcome a fat loss plateau?
5. How does a 5kg increase in lean muscle mass affect daily resting calorie burn?